A TITLE

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Abstract. An abstract

1. Type theory

- Judgemental equality vs propositional equality: $\equiv$ vs $=$
- Function types
    - Definition in closed form or open form

1.1. Universes

1.2. Type families

Functions of type $B : A \to \mathcal{U}$, i.e. for $x : A$, we have $B(x)$ a type.

1.3. Dependent functions

Functions whose output type depends on the input (type or value). Given a type A and a family $B : A \to \mathcal{U}$, write

$$f : \prod_{x:A} B(x)$$

to mean a function which takes a parameter x of type A and returns a value of type $B(x)$.

1.4. Product types

1.4.1. *"Recursor"*

- Generic way to construct functions taking product parameters, using simple functions

$$\begin{aligned} \mathrm{rec}_{A\times B} &: \prod_{C:\mathcal{U}} (A \to B \to C) \to A \times B \to C \\ \mathrm{rec}_{A\times B}(C, g, (a, b)) &:\equiv g(a)(b) \end{aligned}$$

Projections:

$$\begin{aligned} \pi_1 &:\equiv \mathrm{rec}_{A\times B}(A, \lambda a \,.\, \lambda b \,.\, a) \\ \pi_2 &:\equiv \mathrm{rec}_{A\times B}(B, \lambda a \,.\, \lambda b \,.\, b) \end{aligned}$$

References

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